# 2526-MA378: Assignment 1 ANS with outline solutions

Information on the deadline, weighting, collaboration policy, submission process, etc, is on Canvas. Direct link: https://universityofgalway.instructure.com/courses/46941/assignments/140160

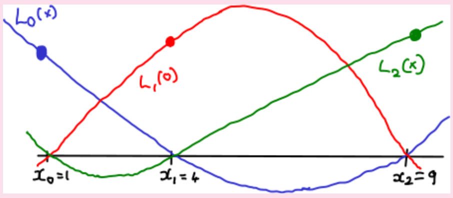
Q1. (Based on the 22/23 exam paper) Let $\{x_0, x_1, x_2\} := \{1, 3, 5\}$ be a set of interpolation points. Write down the formulae for the associated Lagrange Polynomials, $\{L_0, L_1, L_2\}$, and give a rough sketch of them, clearly indicating their values at the interpolation points.

Suppose we define $q(x) := 1 - L_0(x) - L_1(x) - L_2(x)$. Show that, in fact, $q(x) \equiv 0$, for **all** real numbers $x$.

**Answer:** The formulae are

$$L_0(x) = \frac{1}{24}(x-2)(x-4), \; L_1(x) = -\frac{1}{5}(x-1)(x-9), \; L_2(x) = \frac{1}{40}(x-1)(x-4), \tag{1}$$

Sketch:




To show that $q(x) = 0$, one can apply direct calculation. Alternatively, observe that $q$ is a polynomial of degree 2, and $q(1) = q(4) = q(9) = 0$. So $q(x) = 0$ for all $x$.

Q2. Let $f(x) = x^{3/2}$. Give the Lagrange form of $p_2$, the polynomial interpolant to $f$ at at $\{x_0, x_1, x_2\} = \{1, 3, 5\}$. Use Cauchy's Theorem to give an upper bound for $|f(4) - p_2(4)|$. How does this compare with the actual error?

**Answer:** Using the $L_i$ as given in (1), one can state $p_2$ as

$$p_2(x) = f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_0(x) = L_0(x) + 3\sqrt{3}L_1(x) + 5\sqrt{5}L_2(x). \tag{2}$$

(One can also write this as $p_2(x) = (1/8 + 3\sqrt{3}/4 + 5\sqrt{5}/8)x^2 + (-1 + 9\sqrt{3}/2 - 5\sqrt{5}/2)x + 15/8 - 15\sqrt{3}/4 + 15\sqrt{5}/8$, or as $p_2(x) = 0.223504380x^2 + 1.204058694x - 0.427563073$ but that is not necessary).

For the error estimate, we'll use that

$$f(x) - p_2(x) = \frac{f'''(\tau)}{3!}(x-1)(x-3)(x-5).$$

To find the upper bound, first note that $f'''(x) = -3/8x^{-3/2}$ is a negative, increasing function on $[1, 5]$. Hence $|f'''(\tau)| \leqslant |f'''(1)| = 3/8$. Then

$$|f(4) - p_2(4)| \leqslant \frac{3/8}{6}|(4-1)(4-3)(4-5)| = 3/16 \approx \mathbf{0.1875}.$$

The actual error, is $|f(4) - p_2(4)| \approx \mathbf{0.03526}$. So the error was over-estimated by a factor of about 5.

Q3. Let $f(x) = x^{3/2}$ again. Write down the formula for the linear spline, $l$, which interpolates $f$ at $\{x_0, x_1, x_2\} = \{1, 3, 5\}$. Use the relevant theorem from Section 2.1 (Linear Splines) to give an upper bound for $|f(4) - l(4)|$. How does this compare with the actual error?

**Answer:**

$$l(x) = \begin{cases} -\frac{x-3}{2} + 3^{3/2}\frac{x-1}{2} & 1 \leqslant x \leqslant 3 \\ -3^{3/2}\frac{x-5}{2} + 5^{3/2}\frac{x-3}{2} & 3 < x \leqslant 5. \end{cases}$$

The relevant theorem tells us that $\|f - l\|_\infty \leqslant (h^2/8)\|f''\|_\infty$. We compute $f''(x) = 3/(4\sqrt{x})$, which is positive and decreasing on $[1, 5]$, so $\|f''\|_\infty = f''(1) = 3/4$. Also, we have $h = 2$. Therefore the error is bounded as $\|f - l\|_\infty \leqslant (4/8)(3/4) = 3/8 = \mathbf{0.375}$. (Note, you could use Cauchy's theorem more carefully, to get a sharper bound, but it is not required). To get the actual error, we evaluate $f(4) = 8$, and $l(4) = (3^{3/2} + 5^{3/2})/2 \approx 8.1882$. So the actual error is about $\mathbf{0.1882}$. So the estimate and true errors differ by about a factor of 2.